Yongmeng (Celeste) Wu

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Research Interests: Hybrid solvers that combine classical numerical methods with machine learning; reliability of neural operators embedded in iterative solvers; multiscale modeling of composites; differentiable simulation

EDUCATION

Tongji University, Shanghai, China Sep 2023 – Jun 2027
B.Eng. in Engineering Mechanics, *Top-notch Student Training Base in Mechanics*
Average Score: 91.70/100 **Ranking:** Top 5%
Relevant Coursework: Linear Algebra, Numerical Methods, Tensor Analysis, Elasticity & Plasticity, Computational Mechanics, Finite Element Programming, Nonlinear FEA, Mechanics of Composite Materials, Fracture Mechanics, Computational Fluid Dynamics, Random Vibration, Probability & Statistics, AI Algorithms
Scholarships: First-Class (2023–24) and Second-Class (2024–25) Scholarships for Outstanding Undergraduates

RESEARCH EXPERIENCE

Centrum IntelliPhysics Lab, Johns Hopkins University Baltimore, MD, USA
Research Intern, Advisor: Prof. Somdatta Goswami

FE–Neural Operator Coupling via Domain Decomposition Apr 2026 – Present
Project lead; collaborating with Dr. Irina Tezaur (Sandia National Laboratories)

- Built a GPU-accelerated JAX solver coupling finite element and neural operator (Transolver) subdomains via non-overlapping Neumann–Dirichlet domain decomposition.
- Implemented and compared interface accelerators: constant relaxation, Aitken, and IQN-ILS (Anderson-type).
- Validated on elastic benchmarks up to a 3D composite with all 100 fiber subdomains replaced by a neural operator; extended to a J2-plastic bolted joint under cyclic loading.

Multiscale Modeling of Fiber Composites with Neural Operator RVE Surrogates Aug 2026 – Present
Project lead; collaborating with Prof. Satyajit Mojumder (Washington State University)

- Replacing the per-point RVE solve of FE^2 with a neural operator that maps macroscale strain and realistic microstructure (fiber clustering, voids) to homogenized stress, inside a differentiable JAX FEM.
- Ported Moulinec–Suquet FFT homogenization to JAX for batched GPU data generation; CNO is the most accurate architecture in current tests. Extending to path-dependent plasticity.

School of Aerospace Engineering and Applied Mechanics, Tongji University Shanghai, China
Student PI, Shanghai Undergraduate Innovation Program, Advisor: Prof. Yunmei Zhao

Deep Finite Element Method (DFEM) for Heterogeneous Heat Transfer Oct 2025 – Jul 2026

- Extended DFEM (energy minimization over the Galerkin weak form) from linear elasticity to heat conduction.
- Enforced Dirichlet conditions exactly via distance functions, with Fourier features and gradient supervision at material interfaces; relative L_2 error below 0.1% (2D benchmarks) and about 0.3% (3D woven composite).
- Parametric warm-starting decreases prediction time to 1.1–1.6 s (~10× faster than FEM) at ~0.3% error.

First-author manuscript, in revision for resubmission.

PRESENTATION

Poster Presentation, IDIES Annual Symposium, Johns Hopkins University Oct 2026
Plug-in Neural Subdomains: FE–NO Coupling in 3D

Oral Presentation in English, '101 Project' Undergraduate Forum in Mechanics, Hangzhou May 2026
Enhancing Deep Finite Element Method (DFEM) for Heterogeneous Heat Transfer via Exact Boundary Constraints and Gradient Supervision; selected to represent Tongji University

AWARDS

- First Prize (Individual & Team), 6th International Engineering Mechanics Contest (Asian Region) Dec 2025
- Second Prize, 15th National Zhou Peiyuan Mechanics Competition Jul 2025
- First Prize, 16th Chinese Mathematics Competition for College Students Dec 2024

SKILLS

Programming: Python (JAX, PyTorch, FEniCSx), C/C++, MATLAB
Numerical Methods: FEM, domain decomposition, quasi-Newton acceleration, FFT homogenization
Languages: Chinese (native); English (IELTS 7.5; GRE 333: V163, Q170, AW3.5)